

Methods: Overlap

ALMA Thesis Planner (automated pipeline)

0.1 0. Conventions fixed at the start

Adopt a single distance and kinematic reference and use them everywhere: $D = 22$ Mpc ($1'' \approx 107$ pc, so the native $0.268''$ beam ≈ 29 pc and the headline $0.5''$ beam ≈ 54 pc), and $v_{\text{sys}} = 1610$ km/s in the radio LSRK convention ($z = 0.00537$). Every velocity in the thesis is LSRK radio; every offset velocity is relative to a clump's own centroid, never to v_{sys} . Useful conversion to memorize: at 219 GHz, $T_{\text{b}}/I = 1.222 \times 10^6 / (\nu_{\text{GHz}}^2 \theta_{\text{maj}} \theta_{\text{min}})$ K per Jy/beam, i.e. ≈ 102 K (Jy/beam) $^{-1}$ for a $0.5''$ circular beam and ≈ 355 K (Jy/beam) $^{-1}$ at the native $0.268''$ beam. That factor is why we taper: the same flux limit buys roughly $2\times$ better brightness sensitivity at $0.5''$ once the per-beam noise penalty of down-weighting long baselines is paid.

Software: CASA 6.x for retrieval, continuum subtraction, and imaging; Python (astropy, spectral-cube, radio-beam, astrodendro, numpy/scipy, matplotlib, emcee only if needed for the stretch goal) for everything downstream. All analysis lives in one git repository, driven by scripts that read a single YAML configuration file. No parameter is ever typed into a notebook by hand.

0.2 1. Week 1 — scope confirmation and line verification

Query the ALMA archive entry for 2013.1.01041.S and its bibliographic links, and run one ADS search on the project code and on "Antennae SiO HNC O overlap" to confirm no published clump-level SiO/HNCO census exists from this OUS. Record the search strings and date in the repository; the census and the velocity-resolved stack are the deliverable either way, so this step bounds framing, not the work.

In the same week, verify every line identification in Splatalogue at the observed frequency, not the rest frequency. Compute $\nu_{\text{obs}} = \nu_{\text{rest}}/1.00537$ and confirm each falls in a spectral window with at least ± 150 km/s of clean baseline on both sides: SiO(5–4) 217.10498 $\rightarrow$ 216.006 GHz; HNCO($10_{0,10}-9_{0,9}$) 219.79827 $\rightarrow$ 218.624; C¹⁸O(2–1) 219.56036 $\rightarrow$ 218.387; ¹³CO(2–1) 220.39868 $\rightarrow$ 219.221; H₂CO($3_{0,3}-2_{0,2}$) 218.22219 $\rightarrow$ 217.056; CH₃OH($4_{2,2}-3_{1,2}$ A⁺) 218.44006 $\rightarrow$ 217.274. Then do the contaminant sweep: list every catalogued transition within ± 200 km/s of each target line and flag it in the config. Two you will certainly find — HC₃N(24–23) at 218.3247 GHz rest sits only ~ 140 km/s from H₂CO($3_{0,3}-2_{0,2}$), and c-C₃H₂ at 217.822/217.940 GHz rest lands in the same window — so the H₂CO integration window in §6c must be narrow enough to exclude HC₃N, and you will state that width in the config before measuring anything. ¹³CO(2–1) sits near the upper edge of the 217.8–219.7 GHz window; check in the actual header where the band edge falls and how much line-free baseline survives on the high-frequency side. If ¹³CO is edge-truncated, that is the moment to discover it, not in month 3.

0.3 2. Months 1–2 — cubes, with a hard handoff date

Retrieval. Download the QA2 products and the pipeline-calibrated measurement set for ‘uid://A001/X12a/X4e’ via `astroquery.alma`. Inspect the QA2 cubes first: check the delivered channel width, the spectral coverage per window, the restoring beams, and whether the delivered cubes are continuum-subtracted. If the QA2 cubes cover the six lines at usable channel width with sane beams, use them and skip re-imaging; the schedule assumes you will re-image only for the two faint shock lines if the delivered products are unusable.

Continuum subtraction. Identify line-free channels per spectral window from a spatially-averaged spectrum over the bright emission, excluding ± 200 km/s around every line in your Splat-alogue table. Run ‘uvcontsub’ with ‘fitorder=1’ and the line-free channel selection recorded in the config. Keep the continuum-only visibilities.

Line cubes. Image each line with ‘tclean’, ‘specmode=cube’, ‘outframe=LSRK’, ‘gridded=mosaic’ (the observation is a mosaic — do not use ‘standard’), ‘deconvolver=multiscale’ with scales [0, 5, 15, 45] pixels, ‘weighting=briggs’ with ‘robust=0.5’, ‘usemask=auto-multithresh’, ‘pbcor=True’ on the final products, and ‘restoringbeam=common’. Cell $0.05''$; velocity coverage 1300–1900 km/s per line. Produce the headline set with ‘uvtaper’ tuned to give a $0.5''$ circular restoring beam — iterate the taper on a single line first, then apply the same taper to all six. Channel binning: keep ^{13}CO , C^{18}O , and $\text{H}_2\text{CO}/\text{CH}_3\text{OH}$ at 2–3 km/s; bin SiO and HNC to 10 km/s to match the quoted sensitivity. Also produce a native-resolution (robust=0.5, no taper) set for the appendix only.

Continuum image. Make one 1.3 mm mfs image from the 231.4–233.4 and 233.2–235.2 GHz windows using the line-free channels (after excluding the $\text{CH}_3\text{OH}/\text{SO}_2$ candidates you flagged), ‘nterms=1’, ‘gridded=mosaic’, robust=0.5, tapered to the same $0.5''$ beam.

Common grid. Before anything is measured, regrid every cube — headline and appendix sets separately — onto an identical spatial pixel grid and convolve to an identical circular beam using ‘imsmooth’/‘radio-beam’. Verify with an assertion in code that all headers agree on CRVAL, CDELTA, and BMAJ/BMIN/BPA. Aperture photometry across cubes is meaningless otherwise.

Noise characterization. For each cube, build a per-pixel, per-channel noise map: take the flat-noise (non-pbcor) cube, compute the rms in line-free channels per pixel with a median-absolute-deviation estimator, then divide by the primary-beam response map to get the noise map on the pbcor grid. Sanity-check that the flat-noise rms is spatially near-uniform and that the resulting SiO noise at the mosaic center is within a factor of ~ 2 of the 0.975 mJy/beam per 10 km/s scaled by the tapering penalty you measured. Everything downstream — the depth mask, the upper limits, the stacking weights — is read off these maps, never off a predicted sensitivity.

Handoff. Set a calendar date at the end of month 2. Month 3 begins on whatever cubes exist that day. If a line is not imaged by then, it is dropped from the headline analysis and that is recorded.

0.4 3. Month 3 — one clump catalog

Segment the $0.5''$ $^{13}\text{CO}(2-1)$ cube only. Run `astrodendro` on the flat-noise cube with ‘min_value = 3σ ’, ‘min_delta = 2σ ’, and ‘min_npix’ equal to two spectral channels times one beam area in pixels, where σ is the median line-free rms. Keep **leaves only** — no trunk or branch structures enter the catalog. Expect a few tens of objects; if you get hundreds, your min_delta is too permissive and you are cataloguing noise ridges, so check by rerunning on the inverted (negated) cube and

confirming near-zero spurious leaves at your thresholds.

For each leaf, measure CPROPS-style moments on the pbcor cube within the leaf mask: intensity-weighted centroid position and velocity; second moments σ_{maj} , σ_{min} , σ_{v} ; peak and integrated ^{13}CO intensity. Apply the two standard corrections and report both raw and corrected values: extrapolate the moments to the 0 K isosurface (linear extrapolation in the moment-vs-threshold curve), and deconvolve the spatial sizes as $\sigma_{\text{dec}} = \sqrt{(\sigma_{\text{obs}}^2 - \sigma_{\text{beam}}^2)}$, flagging as unresolved anything where $\sigma_{\text{obs}} < 1.1 \sigma_{\text{beam}}$. Convert to a deconvolved radius $R = 1.91 \sigma_{\text{dec}}$ (in pc) and FWHM linewidth $\Delta V = 2.355 \sigma_{\text{v}}$.

Add the **uv-filtering diagnostic** for each clump: on the ^{13}CO moment-0 map, measure the mean surface brightness in an annulus from $1.5R$ to $3R$ and divide it by the clump's peak. Any clump whose annulus mean is more negative than -10% of its peak is flagged 'bowl_flag=True'. These clumps stay in the catalog and in the census; the flag is carried through every table and every figure so the reader can see whether the result depends on them.

Use $\text{C}^{18}\text{O}(2-1)$ as a check, not an input: extract C^{18}O over each ^{13}CO leaf footprint, record the C^{18}O velocity centroid where detected at $\geq 3\sigma$, and record the ratio $I(^{13}\text{CO})/I(\text{C}^{18}\text{O})$. Where that ratio falls well below the assumed abundance ratio (adopt $[^{13}\text{CO}]/[\text{C}^{18}\text{O}] = 8$), flag the clump as optically thick in ^{13}CO — the column derived in §4 is then a lower limit and the clump is annotated as such.

0.5 4. Month 4 — a denominator defined by physics

Two cuts, both fixed in the config before any SiO cube is opened.

Column cut. Convert the ^{13}CO peak brightness temperature and linewidth into a peak H_2 column under LTE with a fixed $T_{\text{ex}} = 20$ K, optically thin ^{13}CO , and CMB background subtraction:

$$N(^{13}\text{CO}) = (8\pi\nu^3/c^3) \cdot [Q(T_{\text{ex}})/(g_{\text{u}} A_{\text{ul}})] \cdot \exp(E_{\text{u}}/kT_{\text{ex}}) \cdot [\exp(h\nu/kT_{\text{ex}}) - 1]^{-1} \cdot T_{\text{R}} dv,$$

with $E_{\text{u}} = 15.87$ K, $A_{\text{ul}} = 6.04 \times 10^{-7} \text{ s}^{-1}$, $g_{\text{u}} = 5$, and $Q(T) \approx 2kT/hB + 1/3$ for $B = 55.101$ GHz. Then $N(\text{H}_2) = N(^{13}\text{CO})/X(^{13}\text{CO})$ with $X(^{13}\text{CO}) = 2 \times 10^{-6}$ (from $\text{CO}/\text{H}_2 = 10^{-4}$ and $^{12}\text{C}/^{13}\text{C} = 50$). Set the cut at a round number safely above your completeness limit — compute the $N(\text{H}_2)$ corresponding to a 5σ ^{13}CO peak at the median linewidth, round up to the next factor-of-two-clean value (this will land near 10^{22} cm^{-2}), and freeze it. Quote every assumption in the same table as the cut.

Depth cut. From the SiO noise map, compute the 3σ integrated-intensity limit per pixel over a fixed 40 km/s window, and define the census footprint as the contiguous region where that limit is below a stated uniform value (choose the value so the footprint covers most of the mosaic's high-response area; read it off the map, do not guess). Only clumps whose ^{13}CO peak position lies inside this footprint enter the census.

Every clump excluded by either cut goes into a separate table with the reason for exclusion. The census fraction is quoted only over the surviving sample.

Pre-registration mechanics. At the end of month 4, commit and tag the config file containing: the column cut, the depth threshold, the integration-window rule, the detection criterion, the stacking weights, the H_2CO ratio threshold, and the inconclusiveness test of §6c. Until that tag

exists, you do not extract a single SiO or HNC spectrum. This is the whole reason the result is quotable.

0.6 5. Months 5–6 — measurement

For every surviving clump, extract SiO and HNC spectra by averaging over the clump’s ^{13}CO leaf mask projected onto the sky (the 2D union of the mask across velocity), on the common-grid, common-beam pbcor cubes. Integrate over a fixed window centered on the clump’s C^{18}O centroid where available, else its ^{13}CO centroid, with half-width $\max(1.5 \times \text{FWHM}(^{13}\text{CO}), 20 \text{ km/s})$, capped at $\pm 40 \text{ km/s}$.

Compute the uncertainty on the integrated intensity as $\sigma_I = \sigma_{\text{chan}} \cdot \Delta v_{\text{chan}} \cdot \sqrt{N_{\text{chan}}} / \sqrt{N_{\text{beam}}}$, where σ_{chan} is the aperture-averaged noise from the noise map, N_{chan} is the number of channels in the window, and N_{beam} is the number of independent beams in the aperture. Verify this empirically rather than trusting it: place several hundred apertures of the same size at random positions inside the depth footprint but off the clumps, integrate over the same window width, and confirm the measured scatter matches σ_I within $\sim 20\%$. If it does not, use the empirical scatter.

Detection criterion (single, pre-registered): $I \geq 3\sigma_I$ in the fixed window. As a confirmation check reported alongside — never as an alternative criterion — require that the detection show ≥ 2 contiguous channels above 2.5σ in the 10 km/s -binned spectrum. Non-detections get a 3σ upper limit, $I_{\text{lim}} = 3\sigma_I$, with the local noise value attached to the catalog row.

0.7 6. Months 7–9 — three results

(a) Column-selected detected fraction. Report $N_{\text{det}}/N_{\text{sample}}$ for SiO, for HNC, and for either-line, each with a Wilson score 95% interval. Every quotation of the fraction appears in the same sentence as the column cut and the limiting depth — write it that way in the abstract, the results, and every figure caption. Report the fraction once more with ‘bowl_flag’ clumps removed, in the same table.

(b) Velocity-resolved stack. Shift each clump’s SiO and HNC spectrum onto a velocity-offset axis defined by its C^{18}O centroid (^{13}CO where C^{18}O is undetected; record which was used per clump), resample onto a common 10 km/s grid spanning $\pm 100 \text{ km/s}$ so the $\pm 30 \text{ km/s}$ offsets the science motivates are well inside the window, and co-add with inverse-variance weights (weights fixed in the config). Produce three stacks: all clumps, non-detections only, and detections only. The headline figure is stacked SiO and HNC intensity versus velocity offset, with the ^{13}CO stack overplotted as the bulk-gas reference, plus a companion panel showing stacked S/N as a function of integration-window width from 10 to 100 km/s .

Run three null tests and show at least the first: stack at the same positions but with each clump’s velocity offset randomized; stack at random off-clump positions inside the depth footprint; and jackknife the sample by leaving out one clump at a time, reporting the spread. Uncertainties on stacked quantities come from bootstrap resampling of the clump list (10^4 draws), not from the propagated σ_I alone.

Report line strengths as **intensity ratios** $I(\text{SiO})/I(^{13}\text{CO})$ and $I(\text{HNC})/I(^{13}\text{CO})$ computed from the stacks. Do not convert to abundances anywhere — one transition per species constrains no excitation. Secondary split: divide the sample into ^{13}CO linewidth quartiles and show the four binned stacks in one panel. No per-clump ratio distributions, no correlation coefficients on limits.

(c) Column-matched selectivity test. Measure $I(\text{H}_2\text{CO } 3_{0,3}-2_{0,2})$ for every clump using a window narrow enough to exclude $\text{HC}_3\text{N}(24-23)$ at $+140$ km/s (state the width; ± 60 km/s is the natural choice). Form $R_{\text{H}_2\text{CO}} = I(\text{H}_2\text{CO})/I(^{13}\text{CO})$ and split the sample at the threshold you froze in month 4 (the sample median of $R_{\text{H}_2\text{CO}}$ computed on ^{13}CO and H_2CO alone, before any SiO spectrum is opened).

Match the two halves in column: bin all clumps in $I(^{13}\text{CO})$ using bins of fixed width in $\log I$, and within each bin keep equal numbers of H_2CO -bright and H_2CO -faint clumps (random subsampling, averaged over 10^3 realizations). Stack SiO within each matched sample and compare $I(\text{SiO})/I(^{13}\text{CO})$ between them, with bootstrap confidence intervals on the difference.

Before interpreting anything, run the pre-registered overlap test: a two-sample KS test on the $I(^{13}\text{CO})$ distributions of the H_2CO -bright and H_2CO -faint halves. If $p < 0.05$, or if fewer than three $I(^{13}\text{CO})$ bins contain at least two clumps of each type, the test is reported as **inconclusive** and the stacks are shown without a claim. Write that sentence into the config file now so you cannot be tempted later.

0.8 7. Continuum

Use the 1.3 mm image for one purpose only: overlay the continuum $\geq 5\sigma$ contours on the clump catalog and set a boolean ‘cont_flag’ for clumps with a coincident continuum peak, noted in the text as “possibly hosting an embedded HII region.” Band 6 alone cannot separate dust from free-free, so no continuum-derived mass, no spectral index, no dust column enters the thesis.

0.9 8. Appendix robustness table (one page, once)

Repeat §6a — the fraction and its Wilson interval — for exactly two variants: the native-0.268'' catalog and cubes (with the depth and column cuts recomputed from that configuration’s own noise map), and a C^{18}O -based column and centroid variant at 0.5''. One table, three rows, no discussion in the main text.

0.10 9. Optional stretch, only if months 7–8 finish early

For the brightest few SiO detections, compute LTE column densities over an assumed T_{ex} grid of 20–100 K and plot $N(\text{SiO})$ versus T_{ex} so the assumption’s effect is visible on the page. This is a figure with a caption, not a result.

0.11 10. Deliverables and reproducibility

The repository ships: the 0.5'' cubes and moment-0/1/2 maps for all six lines plus the continuum image; one machine-readable clump catalog (FITS table and CSV) with position, deconvolved size, velocity centroid, linewidth, peak and integrated ^{13}CO , $\text{N}(\text{H}_2)$, C^{18}O centroid and ratio, H_2CO intensity, $R_{\text{H}_2\text{CO}}$, SiO and HNC integrated intensities or 3σ limits with local noise, ‘bowl_flag’, ‘cont_flag’, and exclusion reasons; the tagged pre-registration config; the three headline figures; and the appendix table. Every figure is regenerated by one script from the catalog and the config, with no manual steps. Run that regeneration from a clean checkout before you submit.